

SEMINAR 5 (Pilot). LIMITS OF FUNCTIONS. COMPUTATIONS

1. Find the domain and the range for the following functions:

$$(a) \ f(x) = \frac{1}{1+x}; \quad (b) \ f(x) = \begin{cases} |1-x|, & x < 2 \\ \sqrt{x+2}, & x \geq 2 \end{cases}$$

2. (HW) Find the domain and the range for the following functions:

$$(a) \ f(x) = \sqrt{2+x-x^2}; \quad (b) \ f(x) = \begin{cases} x+1, & -3 \leq x < -1 \\ 2-x^2, & x \geq -1 \end{cases}$$

3. Use the $\varepsilon - \delta$ definition of limit to prove that $\lim_{x \rightarrow 2} x^2 = 4$.

4. (HW) Use the $\varepsilon - \delta$ definition of limit to prove that $\lim_{x \rightarrow 9} \sqrt{x} = 3$. Find δ such that $|\sqrt{x} - 3| < 0.01$ whenever $0 < |x - 9| < \delta$.

5. Use negation to the Heine definition of limit to show that $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist.

6. (HW) Use negation to the Heine definition of limit to show that $\lim_{x \rightarrow 0} \frac{x}{|x|}$ does not exist.

7. Find the following limits:

$$\begin{aligned} (a) \ \lim_{x \rightarrow 0} \frac{x^2 + 3x}{x}; & \quad (b) \ \lim_{x \rightarrow 2} \frac{4x^2 - 12x + 8}{x^2 + 3x - 10}; & \quad (c) \ \lim_{x \rightarrow 1} \frac{x^3 + 4x^2 - 5}{x^4 - 1}; \\ (d) \ \lim_{x \rightarrow 7} \frac{7-x}{\sqrt{x+9}-4}; & \quad (e) \ \lim_{x \rightarrow 4} \frac{x^2 - 9x + 20}{\sqrt{4x+x^2-16}-x}; & \quad (f) \ \lim_{x \rightarrow \pm\infty} \frac{2 \cdot 5^x - 3}{9 \cdot 5^x + 4}. \end{aligned}$$

8. (HW) Find the following limits:

$$(a) \ \lim_{x \rightarrow a} \frac{x^2 - 1}{2x^2 - x - 1} \text{ for } a = 0; \quad a = 1; \quad a = \infty; \quad (b) \ \lim_{x \rightarrow 0} \frac{(1+x)(1+2x)(1+3x) - 1}{x}.$$

9. (HW) Find the following limits:

$$(a) \ \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x - 1} - \sqrt{x^2 + 7} \right) \quad (b) \ \lim_{x \rightarrow 3} \frac{\sqrt{6+x} - x}{\sqrt{28-x} - 5}.$$

10. (HW) Find the following limits:

$$(a) \ \lim_{x \rightarrow +\infty} \frac{3 \cdot 2^x - 7 \cdot 3^x}{2^{x+2} + 5 \cdot 3^x}; \quad (b) \ \lim_{x \rightarrow -\infty} \frac{3 \cdot 2^x - 7 \cdot 3^x}{2^{x+2} + 5 \cdot 3^x}.$$

11*. (HW) Find $\lim_{x \rightarrow x_0}$ for any $x_0 \in \mathbb{R}$ if

$$f(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \text{ } (x \text{ is rational}), \text{ with } p \in \mathbb{Z} \text{ and } q \in \mathbb{N} \text{ coprime} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$